

Transformer Embeddings

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Model Architecture & Training Ablations

Model Architecture and Refactoring

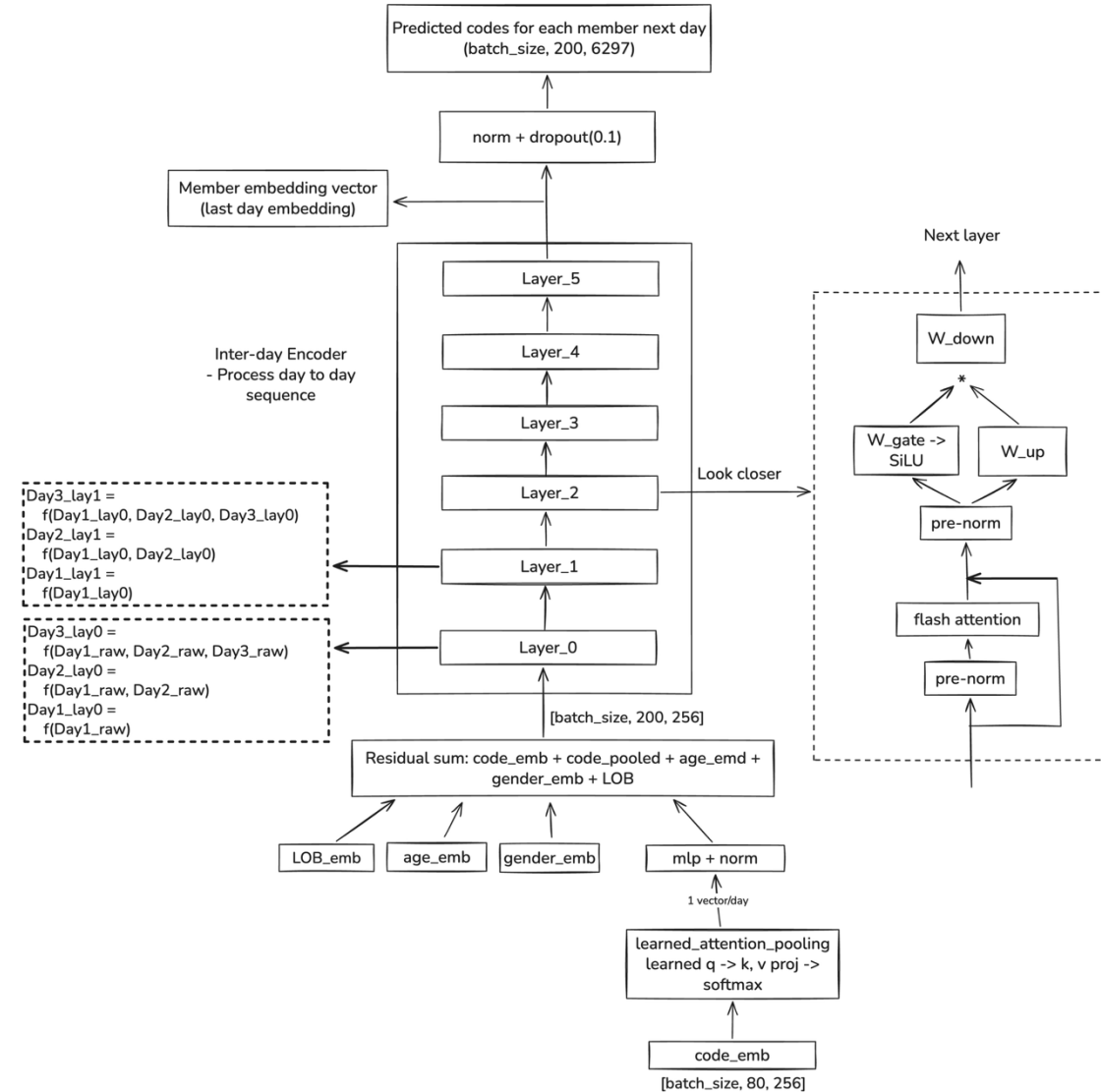
- Preserve hierarchical structure
- Hierarchical structure for encoding within- (80 codes) and inter-day (200 days)
- Trained to predict next-day medical codes (multi-label, 6,297 target groups)

Training & inference efficiency

- Replace the attention layer with **flash attention (FA)** to avoid GPU IO bottleneck and provides faster training with less GPU memory
- Replace the regular FFN with SwiGLU FFN for stable training and context-aware feature selection

Optimize training strategy

- Weighted BCE loss accounting for rare codes
- Enable mixed precision training
- SGD -> Adam learning rate optimizer
- Head dimension 16 -> 32 (fit flash attention)



Ablation Results

- Data
 - Commercial: 1.06M
 - Medicare: 0.56M
 - Medicaid: 0.13M
 - Day count > 10
 - Time period: 1/1/2023 – 12/31/2023
 - Claim lookback period: 36 months with 90d delay
- Full input codes: 75.51k
- Target codes: 6.30k

Ablation config.:

- Legacy baseline: Dense + legacy config
- Exp2: Dense + optimized config (OC)
- Exp3: Dense + OC + flash attention (FA)
- Exp4: MOE + FA + OC
- One epoch
- Infrastructure:
 - GPU: T4 * 4 (16GB HBM each)
 - Machine type: n1-highmem-64
 - Data parallelism applied

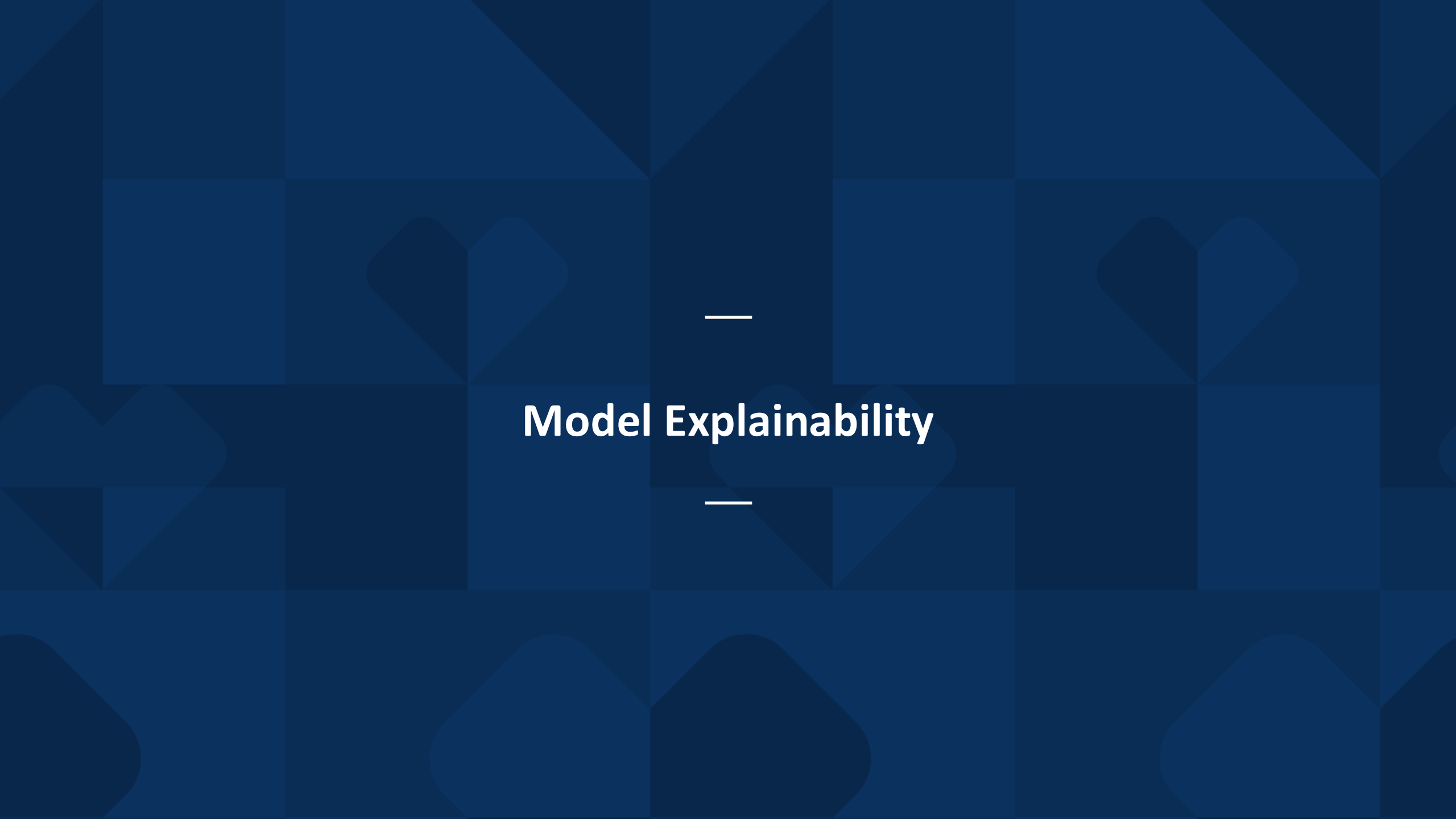
Observations

- Optimized training strategy achieves over **2x** improvement in prediction performance
- Flash attention reduced training cost by **34%** and reduced GPU memory usage by **61%**
- MOE matched dense model performance
- **Notice:** intrinsic metrics captures only internal metrics for training quality and outcomes on predicting next day's labels; **downstream evaluation** provides definite and business-driven measure of the embedding quality.

Ablation config.	Recall@10 ¹	Precision@10 ²	Micro Recall@10 ³	NDCG@10 ⁴	Training time (hrs)	GPU peak memory (GB)
Random reference	0.007	0.001	0.002	0.001	NA	NA
Legacy model (baseline)	0.579	0.120	0.234	0.169	6.15	28.29
Optimized model	0.825	0.235	0.478	0.401	9.65 ⁵	20.58
Optimized + FA	0.828	0.237	0.462	0.398	4.09	11.13
Optimized + FA + MOE	0.827	0.237	0.461	0.399	4.74	13.39

1. Recall@10: fraction of members have at least 1 correct code in top-10 predictions
2. Precision@10: fraction of correct predictions at top-10 predictions
3. Micro-recall@10: fraction of true codes found in top-10 predictions
4. NDCG@10 (normalized discounted cumulative gain at top-10), indicates ranking quality, where 1 is perfect and 0 is none is relevant at top10 predictions
5. Due to the constraints of GPU memory and use of data parallelism, this experiment used 16 batch size instead of 32. The training time, given more advanced GPU, should be less than 6 hours

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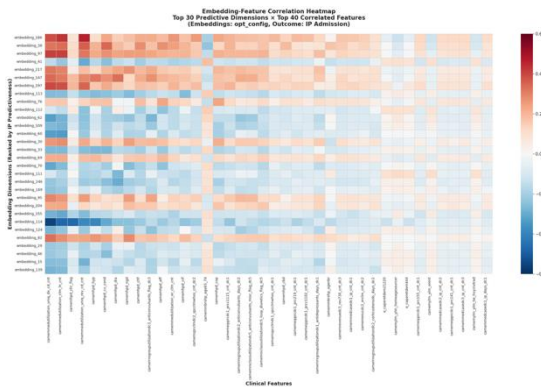
Model Explainability

Review of previous work in TE explainability

Task-oriented

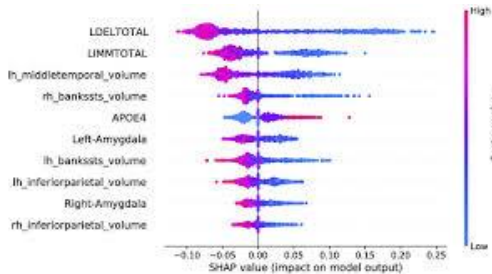
Semantic meanings of the embedding

Correlation between a given TE dimension with traditional features, e.g., emb154 is associated with diabetes in Medicaid; emb186 encodes claims volume and utilization patterns



Primary driver of predictions

Identify **importance of specific embedding dimensions** on downstream classification tasks by measuring effects of removing such dimensions



Primary driver of prediction – LLM

Use **LLM** to directly generate reasoning and explanations of the prediction outcomes, given the raw claim inputs and derived feature importance.

Member A:

Service Date: 2022-12-30
icd9_dx_cd: I10,G93.40
prodr_cd: 84443,82607,82746,84439
hcfa_plc_srv_cd: 81
taxonomy codes: 29IU00000X
...
Service Date: 2023-01-05
icd9_dx_cd: I10,E11.621
prodr_cd: 99309
hcfa_plc_srv_cd: 31
...

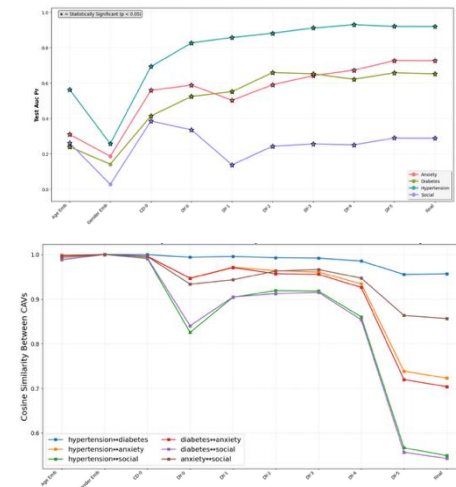


Gemini: “The member faces a high likelihood of inpatient admission in the next three months due to several compounding clinical factors.....”

Task-agnostic (Model behavior)

Model learning dynamics and concept representations

Dissect the transformer model **layer-by-layer** to understand what clinical concepts are learned, where they are encoded, and how the learning information evolves through the network's depth



Semantic meanings of embeddings with correlation

- **Embedding 224 is associated with nearly everything**
 - Most statistically significant association is major chronic count
 - Directionality check shows higher value indicative of greater disease burden
 - *Strong future IP risk associations has been found in all LoBs*
- **Embedding 154 is the top embedding associated with diabetes across LoBs**
 - Directionality check shows that a lower embedding value is indicative of having diabetes (Medicaid shown)

emb154_n <-1	-1<=0	0 to 1	> 1
DIA			
0	0.0%	13.9%	76.0%
1	0.2%	63.8%	35.8%

emb224_n <-1	-1<=0	0 to 1	> 1
major_chronic_cnt			
0	1.6%	76.6%	21.6%
1	0.5%	64.6%	34.6%
2	0.1%	49.1%	50.1%
3	0.0%	35.2%	63.2%
4	0.0%	23.9%	73.1%
5	0.0%	15.6%	78.8%
6	0.0%	9.8%	81.3%
7	0.0%	5.9%	80.3%
8	0.0%	3.7%	76.6%
9	0.0%	2.3%	71.1%
10	0.0%	1.3%	64.7%
11	0.0%	0.8%	57.0%
12	0.0%	0.4%	49.4%
13	0.0%	0.3%	41.0%
14	0.0%	0.2%	34.0%
15	0.0%	0.1%	28.2%
16	0.0%	0.0%	23.8%
17	0.0%	0.0%	18.9%
18	0.0%	0.1%	14.0%
19	0.0%	0.1%	12.3%
20	0.0%	0.0%	8.6%
21	0.0%	0.0%	6.9%
22	0.0%	0.0%	6.8%
23	0.0%	0.0%	4.9%
24	0.0%	0.0%	5.8%
25	0.0%	0.0%	6.4%
26	0.0%	0.0%	0.0%
27	0.0%	0.0%	0.0%
28	0.0%	0.0%	0.0%
29	0.0%	0.0%	0.0%
31	0.0%	0.0%	0.0%

Primary driver of prediction – LLM

Claim history into TE

Service Date: 2022-12-30
icd9_dx_cd: I10,G93.40
prcdr_cd: 84443,82607,82746,84439
hcfa_plc_srv_cd: 81
taxonomy codes: 291U00000X
...
Service Date: 2023-01-05
icd9_dx_cd: I10,E11.621
prcdr_cd: 99309
hcfa_plc_srv_cd: 31
...

+

Other individualized top non-TE features by SHAP

of claim lines in the past 6 months: 20.86,
age: 80.0,
of PCP visits in the past 3 months: 1.0,
of days from recent rx to index_dt: 20.0,
Dementia condition: 1.0
...



**Gemini
Flash 2.5
& prompt**

Reasoning narrative:

The member faces a high likelihood of inpatient admission in the next three months due to several compounding clinical factors.

- **Complex Diabetes with Complications:** Uncontrolled Type 2 Diabetes, evidenced by recurrent foot ulcers, cellulitis, abscesses, and osteomyelitis, presents a continuous risk for severe infections and surgical needs. The recent inpatient stay for these issues underscores this vulnerability.
- **Advanced Cognitive Impairment:** Diagnoses of Alzheimer's disease, metabolic encephalopathy, and altered mental status increase the risk of self-neglect, medication non-adherence, falls, and inability to manage acute symptoms, often necessitating hospitalization.
- **Recurrent Infections and Dehydration:** Episodes of C. difficile enterocolitis and urinary tract infections, coupled with dehydration, indicate a fragile immune system and poor physiological reserve, making the member susceptible to acute medical crises.
- **Multisystemic Disease:** The combination of cardiovascular disease, cognitive decline, and chronic infections creates a complex clinical picture where an exacerbation in one system can quickly lead to decompensation requiring acute hospital care.

Model behavior with probing and CAV – what clinical concepts are learned and where

- **Key clinical concepts are captured well over the architecture**
 - The concept information accumulates over layers and the capability varied across the clinical concepts

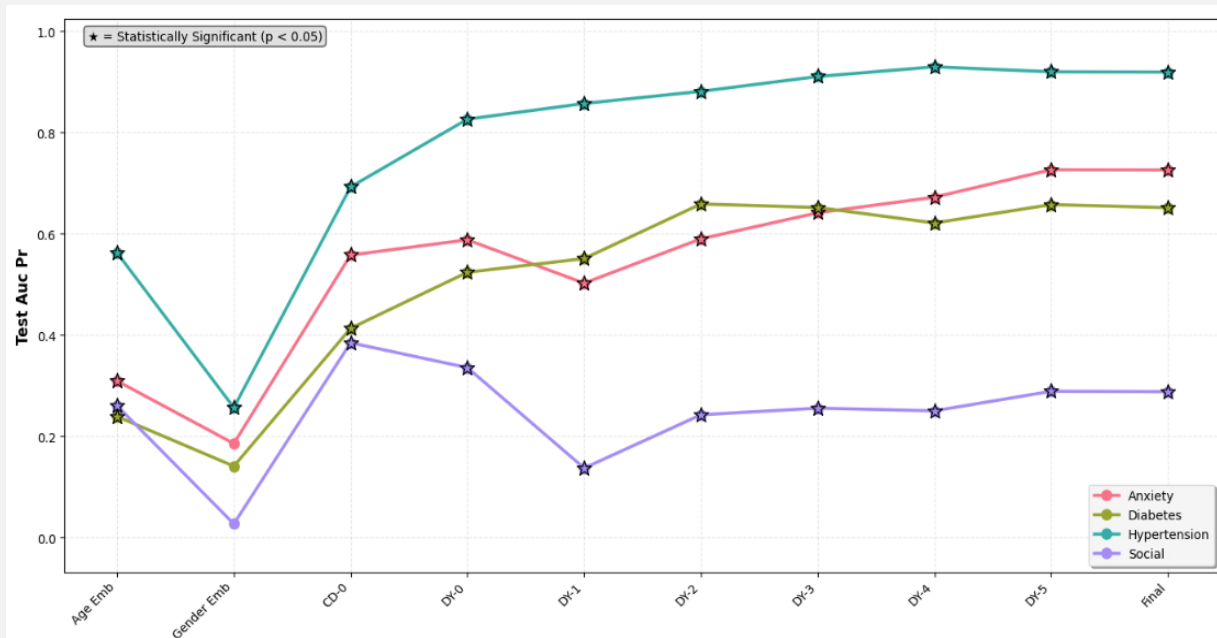


Figure 1: Performance of the transformer encoding different clinical concepts, indicated by probe classifier AUC-PR in predicting the presence of a concept within a Medicaid sampled population.

- **The transformer encodes both shared and specialized information about concepts**
 - It starts to differentiate different clinical concepts approaching the end of layers

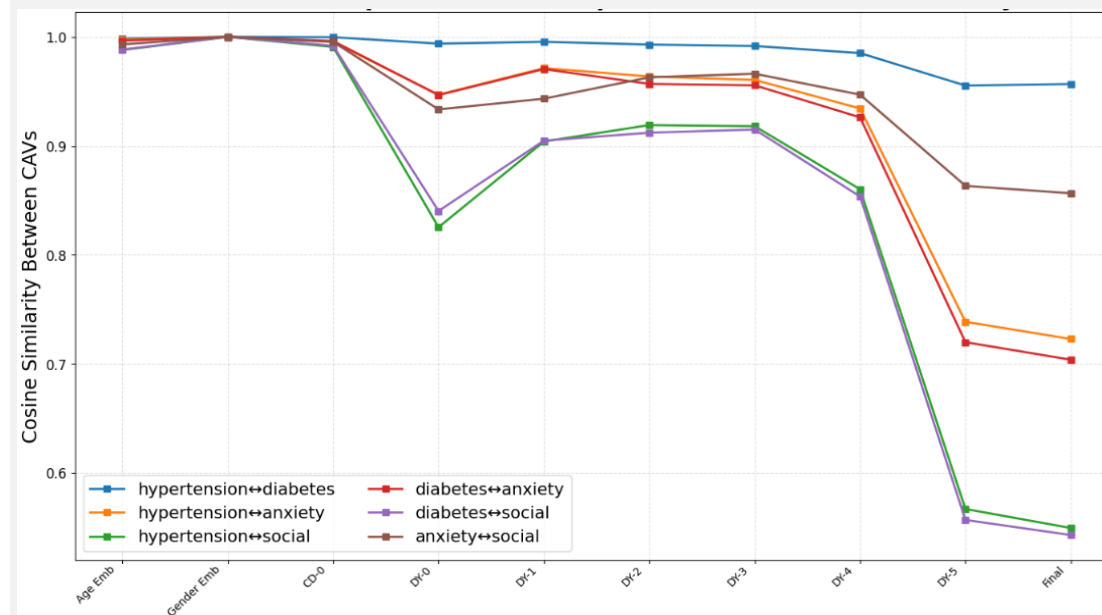


Figure 2: Cosine similarity between clinical concept vectors, indicating how much specialized information each layer encodes for a concept.

TE explainability framework

- What stakeholders really want to understand?
 - Semantic meanings of the embedding dimensions and the impact of each dimension on the downstream classification tasks
 - TE learning progression and how it builds the clinical understanding internally
 - The "explainability" means differently to DS to CM; extra efforts are needed when communicating with CMs.

What principles we should follow?

- **Faithfulness:** The explanation should accurately reflect the models' true behavior, not post-hoc rationalization; and align with domain knowledge and expectations
- **Consistency:** Similar inputs yield similar explanations across contexts (i.e., cross population, upstream-downstream)
- **Transparency:** The transparency should be operated at three levels, i.e., algorithm, procedure and limitations; caution on LLM-based explanation
- **Completeness:** Explanations should capture a sufficient portion of the outputs and model behaviors

What methods we need?

- Simple correlation with traditional features with directional check
- Task-driven methods include Shapley value, LIME
- Model behaviors methods include test with concept activation vector (TCAV), linear probing, attention mechanism visualization
- Experts' generated meaning clinical concepts and ground truth

Method 1 – Concept Activation Vectors (CAV)

- Objective

- Make the embeddings interpretable using Concept Activation Vectors (CAV) to identify which embedding dimensions encode clinically meaningful information

- Method

- Defined 74 claims-based clinical concepts across 9 categories – *Chronic Conditions, Cardiovascular Conditions, Neurological & Mental Health, Cancer & Complex Care, Acute Events, Utilization Patterns, Risk Factors, Care Coordination and others*
- Generated member-level labels from the same data used to train the transformer
- Used CAVs to test which embedding dimensions predict each concept

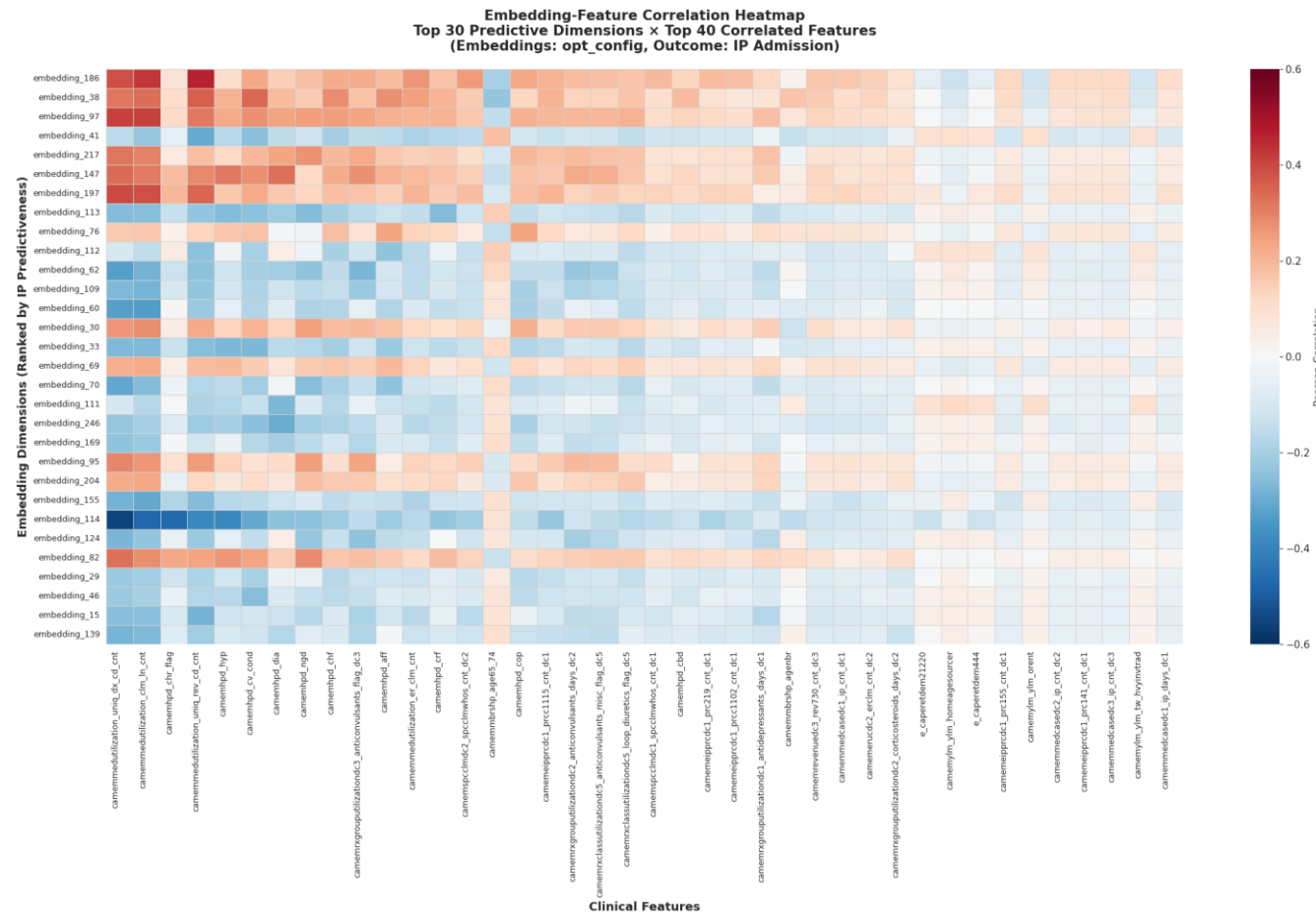
What is CAV?

- Train a linear classifier (logistic regression) to predict each concept from the embeddings
- How to interpret CAV weights:
 - **Positive weight:** Higher dimension value -> Higher probability of concept
 - **Negative weight :** Higher dimension value -> Lower probability of concept
 - **Large absolute weight:** This dimension is important for this concept

Results (Partial)

- **Dim 97 (Importance 0.492) :** Members with higher values have Chronic Kidney Disease, often with metabolic and respiratory comorbidities (obesity, sleep apnea)
- **Dimension 38 (Importance 0.478):** Members experiencing severe acute medical events such as ER use, IP, surgical procedure, myocardial infarction, pneumonia
- **Dimension 52 (Importance 0.423):** Identifies members undergoing cancer treatment or with severe frailty markers like malnutrition, seizures
- **Dimension 189 (Importance 0.397):** Members experiencing sepsis, frequent IP visits and other surgical procedures

Semantic meanings of embeddings with correlation



- **Strong alignment with utilization and disease burden signals**
 - Diagnosis count
 - Total claim lines
 - Revenue code count
 - Chronic condition indicators – Hypertension, Diabetes, CVD
- **Embeddings encode patterns of healthcare interaction**
 - Dimensions reflect overall care intensity and complexity
 - Chronic disease burden (CHF, Hypertension, Diabetes, CV condition) emerge implicitly, which are key risk drivers
- **Embedding_186 (illustrative)**
 - Higher claim volume and diverse service usage
 - Frequent acute utilization (ER visits)
 - Representative of high-touch, more complex members
 - This single dimension captures multi-feature clinical and utilization patterns that would otherwise require many hand-engineered variables

High-Quality, Interpretable Embeddings with Actionable Insights

- Validation : Stability & Robustness

Quality Metric	Value	Interpretation
Directional Consistency (avg sign-rate)	99.9%	Directions are consistent across bootstrap resamples
Mean Stability (std dev)	0.06	Coefficient estimates show low dispersion across samples
Dimensions with clinical meaning	184/256	72% of dimensions have >=1 stable concept association

- Sample Results

Dimension	Clinical Pattern	Example Concepts	Stability
234	Metabolic – Renal	Diabetes (+0.32), Dialysis (-0.31)	100% (2 concepts)
40	SDOH	Financial Strain (+0.13)	99% (1 concept)
211	Cardiovascular	Heart Failure (+0.56)	100% (1 concept)
210	Acute Utilization	ER (+0.91) , Preventable ER (+0.84), Preventative care (-0.32)	100% (7 concepts)

Member-Level Explainability Framework

- **Method**

- **Step 1 – Load member embedding**

- 256-dimensional vector summarizing clinical history

- **Step 2 – Concept Scoring**

- For each clinical concept:
 - Apply trained concept detectors
 - Detect whether the embedding contains signals for that concept
 - Produce a concept score

- **Step 3 – Dimension Attribution**

- Identify which embedding dimensions contributed most to the score
- Contribution = Member Value x Concept Weight

- **Step 4 – Clinical Interpretation**

- Link important dimensions to their meanings using the global dimension dictionary

- **Example: High ER Utilizer**

- **Step 1 – Concept Scores**

- Any ER Visit : 0.92
- Preventable ER Visit: 0.81
- Pneumonia: 0.73
- Poor Care Coordination: 0.58

- **Step 2 – Top Contributing Dimensions**

- Dimension 210: +0.48, Dimension 52: +0.35, Dimension 29: +0.31

- **Step 3 – Clinical Interpretation (LLM-Enhanced)**

- **Example output :**

- *This member demonstrates a high-risk emergency utilization profile, with 92% ED visit likelihood and 81% preventable ED likelihood. The primary driver is Dimension 210, which captures emergency-driven care engagement and inverse associations with preventive behaviors, suggesting reactive rather than proactive healthcare interactions*